

Machine Learning Approaches to Heart Disease Prediction from Clinical Data

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Abstract

Heart disease remains a leading global cause of mortality [1]. This study evaluates 5 machine learning models to predict heart disease using a tabular clinical dataset of 918 patients. After data preprocessing and model comparison, SVM achieved the best overall performance across multiple metrics, including highest ROC-AUC (94.43%). SHAP and permutation importance were applied to interpret the final model. These results demonstrate that machine learning can effectively predict heart disease while maintaining interpretability using structured clinical data.

Introduction

Heart Disease is a major cause of death worldwide and places significant pressure on national healthcare expenditure. In Australia, 42% of all deaths occurred due to cardiovascular disease in 2018 [2]; and in India, from 2005-2015, heart related diseases cost the country around \$237 billion [3]. Therefore, early identification of individuals is important, as timely intervention and preventive management can substantially reduce morbidity and mortality, which would also relieve financial pressure on the public healthcare sector. Machine learning can use routinely collected clinical data to identify potential heart disease patients, who may not be easily detected by the doctors, thereby saving doctors' time on low risk patients and alleviating pressure on healthcare systems. This study will construct and compare 5 machine learning models and evaluate the achievable predictive performance from small sized and low-cost structured data while remaining the model clinically interpretable.

Dataset Analysis

The dataset contains 918 patient records with 11 features used for binary classification of heart disease: Age, Sex, Chest Pain Type, Resting Blood Pressure, Cholesterol, Fasting Blood Sugar level, Resting ECG, Max Heart Rate, Exercise Angina, Old peak, ST Slope¹. Although the size of the data is small, the data integrity is very high; there are no missing values, so no data imputation is needed. Our target variable – Heart Disease is sufficiently balanced, with 55.3% positive and 44.7% negative. The correlation analysis of numerical features shows no evidence of strong multicollinearity, as most pairwise correlations remain below 0.4 in magnitude (Figure 1), suggesting that the feature set is suitable for models that are sensitive to correlated inputs.

¹ The ST segment is the flat, isoelectric section of the ECG between the end of the S wave (the J point) and the beginning of the T wave [13].

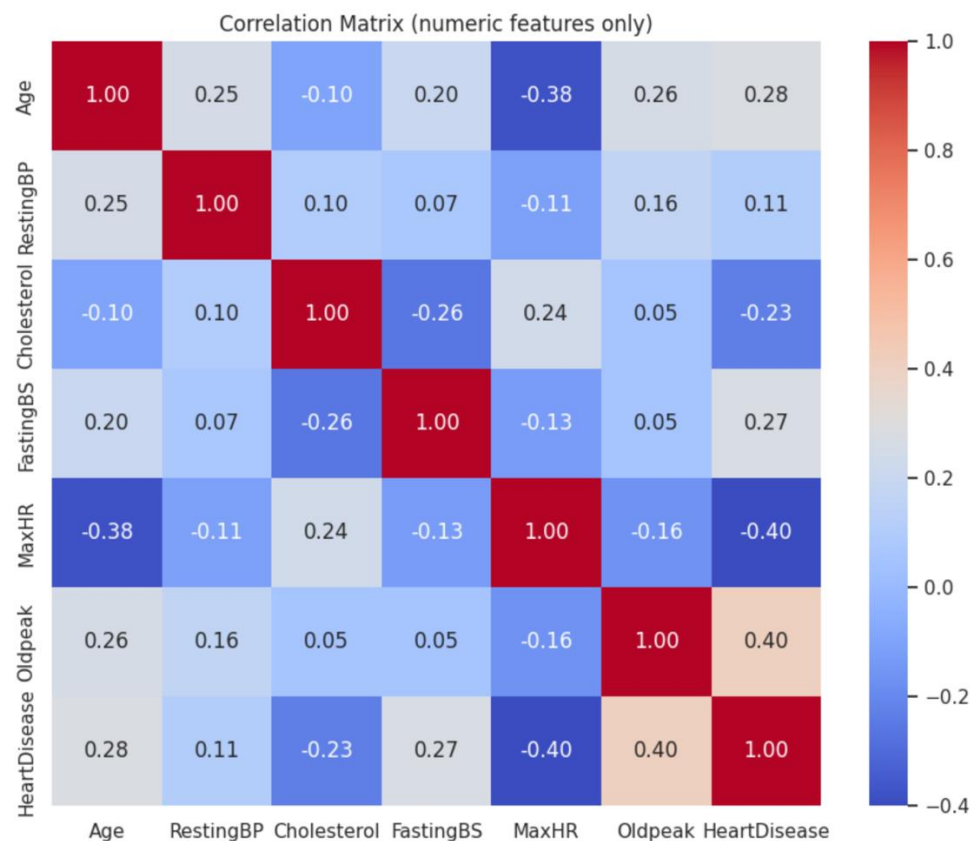


Figure 1 Correlation Matrix of Numerical Features

Methods

Data Preprocessing

The categorical variables (e.g. Chest pain type, resting ECG...) were transformed by one hot encoding to ensure compatibility with the models. Numerical features were standardised using z-score normalisation, to ensure that features with larger scales do not dominate the model training, particularly for distance based models such as SVM. The dataset was divided into an 80% training set and a 20% test set, with stratification applied to preserve the class balance across splits.

Machine Learning Models

Our goal is to construct supervised binary classification ML models to predict whether or not a patient has heart disease. The following models are selected:

Baseline: Logistic Regression

Logistic Regression is chosen to be the baseline model, because it is simply structured and widely used in the healthcare settings, in some studies it even demonstrated better performance than many other non-linear ML models such as Random Forests and XGBoost in the prediction of heart failure [4] [5]. It assumes a linear relationship between the input feature data and their corresponding outputs,

which is highly interpretable. It also acts as a reference when we analyse more complicated models such as MLP, to see if they could offer a better performance over this standard approach.

Random Forest

Random Forest can model the non-linear patterns, and it demonstrated particularly well on small datasets and features, which is very suitable for our dataset. For example, in a similar study of survival prediction with heart failure patients, Random Forest achieved the highest accuracy among other models with a dataset of only 299 patients and 2 features [6];

MLP

MLP is chosen to test if a neural network could offer better performance than other traditional models. Similar research on choosing the best model to predict the binary classification problem of heart failure readmission or death has shown that MLP has the best performance among other ML models [7].

SVM

We chose the SVM with RBF (Radial Basis Function) kernel, this variant of SVM is very powerful for classification problems, as it projects the input features into a higher dimensional space, which is very suitable for complex and high dimensional data [8].

XGBoost

Finally, the XGBoost (eXtreme Gradient Boosting) is selected, as it is known to be able to provide good results in the classification problem [9].

Evaluation

Choice of Metrics

Accuracy measures the overall correctness. To distinguish between different types of errors, I also applied Precision, Recall, Specificity, F1 Score, Receiver Operating Characteristic Area Under the Curve (ROC-AUC), Precision Recall AUC (PR-AUC) and Matthews Correlation Coefficient (MCC). The ROC-AUC is the area under the plot that illustrates our binary classification model at varying thresholds, it is commonly applied in the assessment of diagnostic test performance in clinical epidemiology, whereas accuracy only evaluates the performance at a single threshold. Therefore, we treat ROC-AUC as the most important metric, and rank the results based on the ROC-AUC, with the F1 Score in second place as it balances both Precision and Recall.

Model Performance

	Model	ROC-AUC	F1-score	Accuracy	Precision	Recall (TPR)	Specificity (TNR)	PR-AUC	MCC
0	SVM (RBF)	0.944285	0.915094	0.902174	0.881818	0.950980	0.841463	0.948624	0.803273
1	MLP	0.941176	0.893204	0.880435	0.884615	0.901961	0.853659	0.938478	0.757615
2	Random Forest	0.932209	0.887805	0.875000	0.883495	0.892157	0.853659	0.924998	0.746753
3	XGBoost	0.931612	0.875622	0.864130	0.888889	0.862745	0.865854	0.936410	0.726388
4	Logistic Regression	0.929699	0.900474	0.885870	0.871560	0.931373	0.829268	0.936103	0.769383

Figure 2 Model results for heart disease prediction

All models achieved better performance than random chance (Figure 2).

SVM achieved the best results overall, it has the highest ROC-AUC of 94.43%; and also the highest F1 score of 91.51%, which is notably higher than the second best model (89.32%). It also has the highest Accuracy, Recall, PR-AUC and MCC.

MLP is the second best model, the limitation is likely due to the small dataset. However, the result still demonstrates the great potential of MLP. When we have access to a larger dataset in the future, it would be worth further investigation.

The tree models (Random Forest and XGBoost) have relatively poor performance, but they are still better than our baseline model (Logistic Regression).

Based on our results, we take SVM to be our best model and conduct further investigation. SHAP and Permutation Importance showed us how the prediction was made based on each feature (Figure 3, Figure 4, Figure 5), which suggests that SVM does not only rely on a small set of dominant features, each feature contributes to the final prediction of heart disease.

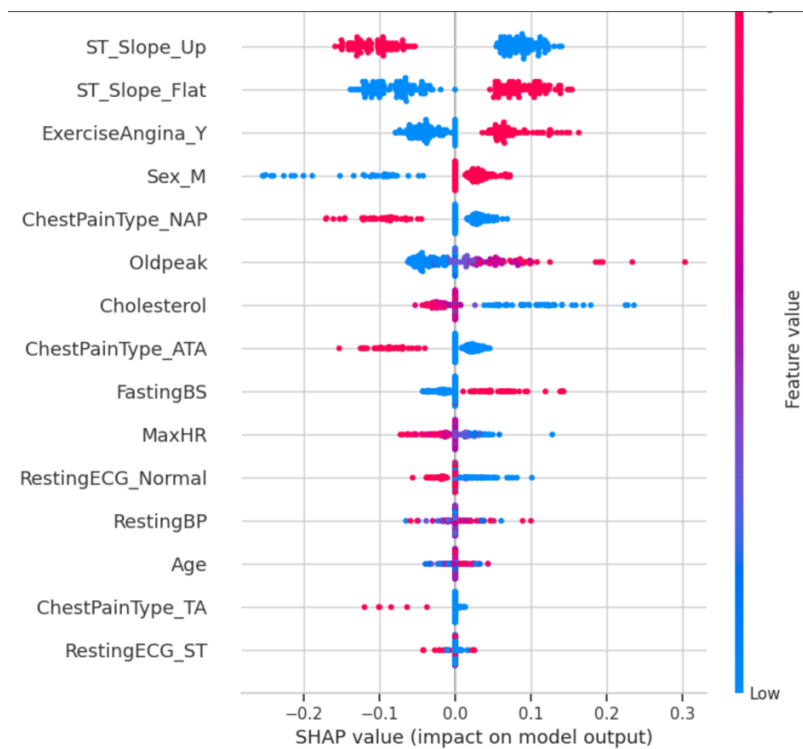


Figure 3 SHAP of the SVM model

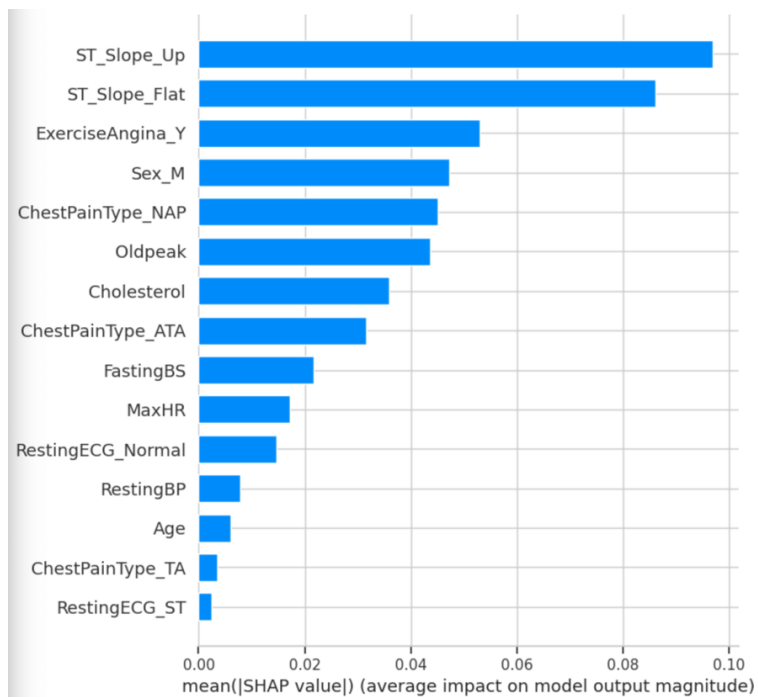


Figure 4 Average impact (mean | SHAP value |) on SVM output

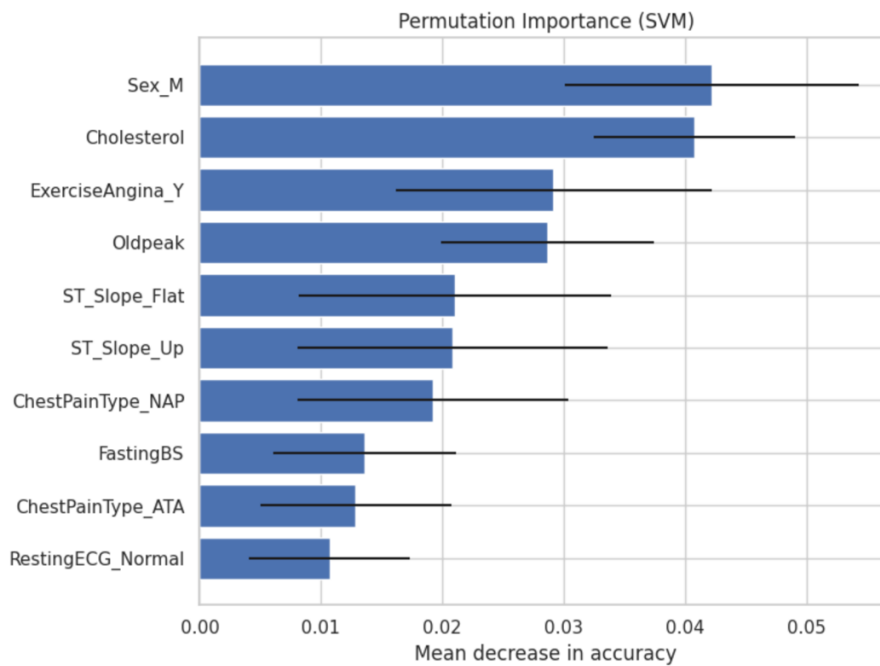


Figure 5 Permutation importance of SVM

Model Limitations and Room for Improvements

Different sensitive attributes may introduce bias. In our dataset, sex isn't evenly distributed, as there are 78.9% males but only 21.02% females. After applying the SVM to the test data based on sex (Figure 6), we can see a clear decrease in specificity (76%), this suggests that there is a higher rate of false positives for male patients; On the other hand, female patients have a relatively lower Recall, which implies a greater risk of false negatives. There is still room of improvement in these issues, one possible approach is to construct the models for males and females individually, so we can train on the data separately, but this would increase the cost of maintenance and deployment, and may also raise disputes regarding fairness and ethical considerations. A more practical approach could be to add more female samples to the dataset, to make the distribution more balanced; or to adjust the threshold parameter in the model for different sexes.

Another major sensitive attribute in our dataset is age. Take 45 years old to be the threshold, 80% of the patients are above 45, therefore it is necessary to examine the performance of SVM by age (Figure 7). Indeed, we see that the precision of SVM for patients who are aged under 45 has dropped to only 66.67%, this implies that roughly one third of the positive predictions are false positives. From a clinical perspective this result is somewhat acceptable as prioritising sensitivity in younger patients reduces the risk of missed diagnoses, but this will lead to unnecessary follow up of healthy patients and therefore an increase in healthcare costs. Similar to the sex based case, we can make improvements by fine tuning the threshold parameters, alternatively, external data improvements such as information on family

disease history and patients' lifestyle could be added as new features for the model to learn from.

	Group	n	ROC-AUC	F1-score	Accuracy	Precision	Recall (TPR)	Specificity (TNR)
0	Male	146	0.920625	0.920000	0.890411	0.884615	0.958333	0.76000
1	Female	38	0.989583	0.833333	0.947368	0.833333	0.833333	0.96875

Figure 6 SVM performance on different sex

	Group	n	ROC-AUC	F1-score	Accuracy	Precision	Recall (TPR)	Specificity (TNR)
0	Age < 45	40	0.964844	0.800000	0.900000	0.666667	1.000000	0.875
1	Age ≥ 45	144	0.938936	0.927083	0.902778	0.908163	0.946809	0.820

Figure 7 SVM performance on age above and below 45

Discussion

The first challenge for this model is the sensitive attributes across different ethnicities. According to a study from the UK Biobank, South Asians have a higher risk of cardiovascular disease compared to white European individuals [10]. Our model was not trained on a dataset that has information about ethnicity, therefore if we deploy such a model in a multi-ethnic country like the UK, this may lead to dataset shift, which would result in biased predictions. A potential solution is to first train a robust model on the majority population, and then apply the model to the minority groups through transfer learning, fine tuning the parameters based on the new data [11].

The second challenge is that the model was evaluated on retrospective data only, without the prospective evaluation, therefore, we do not know how the model is going to perform in the real clinical practice. Studies show that model performance in the real world deployment is often lower than the retrospective evaluations [12], this is because in our retrospective dataset, ground truth already exists, whereas in the real clinical settings, doctors must make decisions under conditions of incomplete information and uncertainty. To solve this problem, the model should undergo prospective evaluation prior to the deployment in the hospitals, so that model predictions can be compared with doctors' decisions, and later evaluated against the true outcomes when they are available.

The third challenge concerns the ongoing maintenance and updating of the model after the deployment under privacy constraints. Due to data protection, patients' data

usually cannot leave the hospitals, therefore a model deployed in a local hospital can only be updated using local data, this would limit its exposure to larger scale and more diverse populations. A solution to this could be federated learning, which enables the model to update across multiple hospitals while keeping the data within each hospital.

Conclusion

This study investigated the feasibility of predicting heart disease using machine learning models trained on tabular clinical data, and compared the performance of five models. SVM achieved the best results and demonstrated its good interpretability through SHAP and permutation importance analysis. The current study still has room for improvement, for example the small data size (918 patients), a larger dataset would permit us to obtain a more reliable result. Additional information on patients such as weight and medical history may also improve our model's performance. Although the model achieved a relatively good result in retrospective evaluation, there are still challenges for real world deployment, such as dataset shift, lack of prospective evaluation, and privacy constraints on model maintenance; approaches such as transfer learning, prospective evaluation and federated learning may help to address these challenges.

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The code used in this study is available at:

<https://github.com/luha-Von-Orius/Machine-Learning-Approaches-to-Heart-Disease-Prediction-from-Clinical-Data.git>